

7

NUMBER AND ALGEBRA

EQUATIONS

Equation-solving has been recorded as far back as 1500 BCE. It was first used in ancient Babylon and Egypt and was brought to Europe from India by the Arabs during the 9th century. The word 'algebra' comes from the Arabic word *al-jabr*, meaning restoration, the process of performing the same operation on both sides of an equation to solve the equation.



Chapter outline

	Working mathematically				
7.01 Equations with variables on both sides	U	F		R	
7.02 Equations with brackets	U	F		R	C
7.03 Equation problems	U	F	PS	R	C
7.04 Equations with algebraic fractions	U	F		R	
7.05 Simple quadratic equations $ax^2 = c$	U	F		R	C
7.06 Quadratic equations $x^2 + bx + c = 0$*	U	F			
7.07 Simple cubic equations $ax^3 = c$*	U	F		R	C
7.08 Equations and formulas	U	F	PS	R	
7.09 Changing the subject of a formula*	U	F		R	C

*** PATH**

Wordbank

cubic equation An equation involving a variable cubed (power of 3), such as $2x^3 = 250$.

equation A mathematical statement that 2 quantities are equal, involving algebraic expressions and an equals sign (=)

formula A rule written as an algebraic equation, using variables.

inverse operation An opposite used in solving an equation, for example, the inverse operation of multiplying is dividing

linear equation An equation involving a variable that is not raised to a power, such as $2x + 9 = 17$.

quadratic equation An equation involving a variable squared (power of 2), such as $3x^2 - 6 = 69$.

solution The answer to an equation or problem, the correct value(s) of the variable that makes an equation true

solve (an equation) To find the value of an unknown variable in an equation

Quiz
Wordbank 7

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (13):

7.01 Two-step equations 1 • Two-step equations 2 • Equations with variables on both sides

7.02 Solving equations • Equations with brackets

7.03 Equation problems

7.04 Solving equations with fractions • Equations with algebraic fractions

7.05 Simple quadratic equations $ax^2 = c$

7.06 Quadratic equations by factorising

7.08 Equations and formulas 1 • Equations and formulas 2

7.09 Changing the subject of a formula

Twig video (1):

7.01 The Arabic science of balancing

PhET interactives (2):

7.01 Equality explorer: Basics • Equality explorer

Quizzes (6):

- Wordbank 7
- SkillCheck 7
- Mental skills 7A
- Mental skills 7B
- Language of maths 7
- Test yourself 7

Skillsheets (3):

SkillCheck Solving equations by balancing • Solving equations by backtracking • Solving equations using diagrams

Worksheets (9):

7.02 Equations 2 • Checking solutions

7.03 Word problems with equations • Angle problems with algebra

7.04 Equations 3

7.05 Equations review

7.08 Working with formulas

7.09 Changing the subject of a formula

Mind map Equations (Advanced)

Puzzles (9):

SkillCheck Solving equations • Backtracking

7.01 Equations with unknowns on both sides

7.02 Equations • Equations order activity

7.03 Writing and solving equations

7.04 Equations code puzzle

7.09 Formulas squaresaw

Language of maths Equations crossword

Presentation (1):

7.03 Applying linear equations

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- solve linear equations
- solve problems involving equations
- (PATH) solve linear equations involving algebraic fractions
- solve simple quadratic equations of the form $ax^2 = c$
- (PATH) solve quadratic equations of the form $x^2 + bx + c = 0$
- (PATH) solve simple cubic equations of the form $ax^3 = c$
- use formulas to solve problems
- (PATH) change the subject of a formula

SkillCheck ANSWERS ON P. 631

1 Solve each **equation**. Use substitution to check your **solutions**.

a $3x - 10 = 5$ **b** $4 + 2y = 21$ **c** $12y + 5 = 23$ **d** $\frac{m}{4} - 6 = 2$

e $\frac{x-5}{4} = 7$ **f** $\frac{5r}{3} = 10$ **g** $\frac{-2x}{5} = 3$ **h** $11 - 2a = 17$

i $20 + 4d = -6$ **j** $\frac{w}{2} + 3 = 4$ **k** $\frac{a}{6} - 1 = 7$ **l** $6 - \frac{n}{3} = 4$

2 Using n to represent 'the number', write an expression for each of these statements.

a The product of the number and 7.
b The square of the number.
c 5 times the sum of the number and 8.
d The number decreased by 20.
e The product of 6 and the number, decreased by 9.
f If the number is even, the *next* even number.

3 Factorise each expression.

a $x^2 + 10x + 21$ **b** $a^2 + 12a - 45$ **c** $y^2 + 24y$

Quiz SkillCheck 7

Puzzles Solving equations

Backtracking

Skillsheets Solving equations by balancing

Solving equations by backtracking

Solving equations using diagrams

7.01

Equations with variables on both sides 7.01

Equations with variables on both sides

For **equations with variables on both sides**, perform operations on both sides to move:

- all the variables onto one side of the equation
- all the numbers onto the other side of the equation.

Puzzles Equations with unknowns on both sides

Interactives Equality explorer: Basics

Equality explorer

Videos The Arabic science of balancing

Two-step equations 1

Two-step equations 2

Equations with variables on both sides

Example 1

Solve each equation.

a $7x + 7 = 2x + 2$ **b** $9 - 6y = 10 - 2y$

Solution

a

$7x + 7 = 2x + 2$

$7x - 2x + 7 = 2x - 2x + 2$ Subtracting $2x$ from both sides

$5x + 7 = 2$

$5x + 7 - 7 = 2 - 7$ Subtracting 7 from both sides

$5x = -5$

$\frac{5x}{5} = \frac{-5}{5}$ Dividing both sides by 5

$x = -1$

Check:

LHS $= 7 \times (-1) + 7 = 0$

RHS $= 2 \times (-1) + 2 = 0$

LHS = RHS.

LHS = left-hand side, RHS = right-hand side.

b

$9 - 6y = 10 - 2y$

$9 - 6y + 2y = 10 - 2y + 2y$ Adding $2y$ to both sides

$9 - 4y = 10$

$9 - 4y - 9 = 10 - 9$ Subtracting 9 from both sides

$-4y = 1$

$\frac{-4y}{-4} = \frac{1}{-4}$ Dividing both sides by (-4)

$y = -\frac{1}{4}$

Check:

LHS $= 9 - 6 \times \left(-\frac{1}{4}\right) = 10\frac{1}{2}$

RHS $= 10 - 2 \times \left(-\frac{1}{4}\right) = 10\frac{1}{2}$

LHS = RHS.

EXERCISE 7.01 ANSWERS ON P. 632

Equations with variables on both sides U F R

EXAMPLE 1

1 Solve each equation and check your solutions. R

a	$5w + 3 = 2w + 21$	**b**	$2q - 10 = q - 4$	**c**	$13x + 1 = 8x + 26$
d	$12n + 3 = 5n - 11$	**e**	$8y - 10 = 10y - 30$	**f**	$3m - 2 = 10 - 3m$
g	$9 - 2a = a - 9$	**h**	$9 - 2x = 18 + 7x$	**i**	$12y + 6 = 6 + 9y$
j	$-12 - 10u = -20 - 18u$	**k**	$15 - 7x = 22 - 3x$	**l**	$10 - 6x = -15 - 11x$

2 For each equation, select the correct solution **A**, **B**, **C** or **D**.

a $6x - 1 = 2x + 11$

A $x = 12$ **B** $x = 3$ **C** $x = 0$ **D** $x = 2.5$

b $11 - 4p = 2p + 2$

A $p = 6.5$ **B** $p = 2$ **C** $p = 1.5$ **D** $p = 3$

3 Solve each equation. R

a	$7w + 15 = w + 3$	**b**	$10 - 3t = 16 + t$	**c**	$4a + 2 = 10 - 4a$
d	$50 + 7y = 20 - 3y$	**e**	$8y - 2 = 10y + 1$	**f**	$9y + 3 = 9 - y$
g	$9 - t = 7t - 2$	**h**	$5y + 2 = 17 - y$	**i**	$25 - 12k = 15 - 6k$

4 Solve $-3n - 8 = -7n - 12$. Select **A**, **B**, **C** or **D**.

A $n = 5$ **B** $n = 2$ **C** $n = -1$ **D** $n = -0.4$

Foundation Standard Complex

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Did you know?

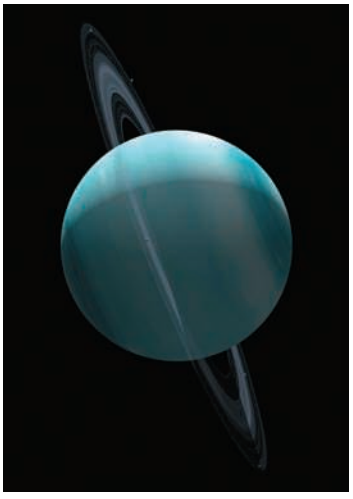
Uranus and Neptune

In 1781, British astronomer William Herschel discovered the planet Uranus (pictured). At that time, it was the farthest planet known in our solar system. However, astronomers found that Uranus' orbit around the Sun did not follow the expected path.

Working separately, mathematicians John Couch Adams of England and Urbain Leverrier of France both predicted that this different orbit was caused by an unknown planet. They calculated the position of this undiscovered planet using a number of equations.

In 1846, a German astronomer called Johann Galle located this planet and named it Neptune.

How long is one day on Uranus?



Science Photo Library/Claus Lunau

Equations with brackets 7.02

Equations with brackets

For **equations with brackets (grouping symbols)**, expand the expressions and then solve as usual.

Example 2

Solve each equation.

a $3(a + 7) = 6$ **b** $9(m - 5) = 7(m + 1)$ **c** $10y - 3(2y - 5) = 6(8 - 3y)$

Solution

a

$3(a + 7) = 6$

$3a + 21 = 6$ Can you think of another way to solve this equation without expanding?

Expanding the expression to make it a 2-step equation

$3a + 21 - 21 = 6 - 21$ Subtract 21 from both sides

$3a = -15$

$\frac{3a}{3} = \frac{-15}{3}$ Divide both sides by 3

$a = -5$ Check: $3(-5 + 7) = 3 \times 2 = 6$

Worksheets Equations 2
Checking solutions
Puzzle Equations
Puzzle Equations order activity
Videos Equations with brackets
Solving equations

b $9(m-5) = 7(m+1)$

$9m - 45 = 7m + 7$ Expanding brackets on both sides

$9m - 7m - 45 = 7m - 7m + 7$ Subtracting $7m$ from both sides

$2m - 45 = 7$

$2m - 45 + 45 = 7 + 45$ Adding 45 to both sides

$2m = 52$

$\frac{2m}{2} = \frac{52}{2}$ Dividing both sides by 2

$m = 26$

Check:
LHS $= 9 \times (26 - 5) = 9 \times 21 = 189$
RHS $= 7 \times (26 + 1) = 7 \times 27 = 189$
LHS = RHS.

c $10y - 3(2y - 5) = 6(8 - 3y)$

$10y - 6y + 15 = 48 - 18y$ Expanding brackets on both sides

$4y + 15 = 48 - 18y$ Collecting like terms

$4y + 18y + 15 = 48 - 18y + 18y$ Adding $18y$ to both sides

$22y + 15 = 48$

$22y + 15 - 15 = 48 - 15$ Subtracting 15 from both sides

$22y = 33$

$\frac{22y}{22} = \frac{33}{22}$ Dividing both sides by 22

$y = 1\frac{1}{2}$

EXERCISE 7.02 ANSWERS ON P. 632

Equations with brackets U F R C

EXAMPLE 2

1 Solve each equation. R

a $2(m + 3) = 8$ **b** $3(x + 1) = 9$ **c** $5(y - 2) = 15y$
d $35 = 7(k + 1)$ **e** $4(3 - a) = 16$ **f** $11 = 9(1 + 2p)$
g $3h = 4(h + 6)$ **h** $6(m - 10) = -6$ **i** $8u = 11(u - 3)$
j $27 = 7(2y + 1)$ **k** $5(2 + 3p) = -8$ **l** $22x = 9(4x - 3)$

2 In which line has an error been made in solving $5(x - 3) = 25$? Select **A**, **B**, **C** or **D**.

$5(x - 3) = 25$

$5x - 8 = 25$ **A**

$5x - 8 + 8 = 25 + 8$ **B**

$5x = 33$ **C**

$x = \frac{33}{5}$ **D**

$= 6\frac{3}{5}$

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3 Show that $k = 5$ is the solution to $12(k - 1) = 48$. R C

4 Show that $a = 6$ is the solution to $10 + a = 2(2 + a)$. R C

5 Solve each equation. R

a	$8(m + 2) = 5(m + 5)$	**b**	$2(y - 3) = 4(y - 5)$	**c**	$3(2 + x) = 4(1 + x)$
d	$5(p + 2) = 3(6 + p)$	**e**	$5n + 6 = 2(2n + 1)$	**f**	$2(4 - 3x) = 4(7 - 3x)$
g	$4(3w - 1) = 5(4 + 3w)$	**h**	$-2(x + 1) = 16 - 5x$	**i**	$-8y - 5 = 5(2y - 3)$

6 Show that the solution to $5(2m - 2) = 6(m + 1)$ is $m = 4$. R C

7 Solve each equation. R

a	$5(m + 6) + 10 = 3(m + 2) + 20$	**b**	$3(y + 2) - 10 = 2(y - 1) + 5$
c	$7y + 2(y + 5) = 4(y - 10)$	**d**	$3x + 4(5 + x) = 6(2 + x) + 20$
e	$5y + 2(y - 3) = 4y + 2(2y + 10)$	**f**	$11 - 2(5 + y) = 4(3 + y) - 1$
g	$8 - 3(1 - m) = 5(m + 3) + 4$	**h**	$12 - 7(2y - 5) = 6 - 15(2 - 5y)$

Equation problems 7.03

Steps in solving word problems using equations

1 Read the problem carefully and determine what needs to be found: 'What is the question?'
2 Use a variable to represent the unknown quantity.
3 Write the problem as an equation.
4 Solve the equation.
5 Answer the problem.

Worksheets
Word problems with equations
Angle problems with algebra

Puzzle
Writing and solving equations

Example 3

When three-quarters of a number is decreased by 8, the result is 46. What is the number?

Solution

Let the number be x.

$\frac{3x}{4} - 8 = 46$ Translating from words to algebra

$\frac{3x}{4} - 8 + 8 = 46 + 8$ Adding 8 to both sides

$\frac{3x}{4} = 54$

$\frac{3x}{4} \times 4 = 54 \times 4$ Multiplying both sides by 4

$3x = 216$

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$\frac{3x}{3} = \frac{216}{3}$ Dividing both sides by 3

$x = 72$ Check: $\frac{3 \times 72}{4} - 8 = 54 - 8 = 46$

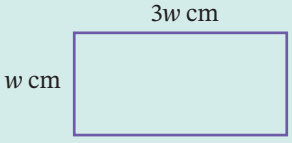
The number is 72.

Example 4

A rectangle is 3 times as long as it is wide. If its perimeter is 60 cm, find its dimensions.

Solution

Let the width of the rectangle be w cm. Then the length is $3w$ cm.



The perimeter is $w + 3w + w + 3w$ and this is given as 60.

$w + 3w + w + 3w = 60$

$8w = 60$

$w = 7.5$

∴ The width of the rectangle is 7.5 cm and the length is $3 \times 7.5 = 22.5$ cm.

Check: The perimeter of a rectangle with dimensions 7.5 cm and 22.5 cm is $7.5 + 22.5 + 7.5 + 22.5 = 60$ cm.

Example 5

The sum of 3 consecutive numbers is 150. Find the numbers.

Solution

Let the first number be x.

The next number is $x + 1$ and the third number is $x + 2$.

Consecutive numbers follow each other in order, such as 7, 8, 9.

Their sum is $x + (x + 1) + (x + 2)$ and this is given as 150.

$x + x + 1 + x + 2 = 150$

$3x + 3 = 150$

$3x = 147$

$x = \frac{147}{3}$

$= 49$

∴ The consecutive numbers are 49, 50 and 51. Check: $49 + 50 + 51 = 150$.

Example 6

Ryan is 6 years older than his sister Kelsey. Their mother is 3 times Ryan's age.

a If the sum of the 3 ages is 59, write a simplified equation to find Ryan's age.

b Solve the equation and find each person's age.



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Solution

a Let x = Ryan's age.
Kelsey's age is $x - 6$. Kelsey is 6 years younger than Ryan
The mother's age is $3x$.

$$x + (x - 6) + 3x = 59$$
$$5x - 6 = 59$$ Simplify the equation

b
$$5x - 6 = 59$$
$$5x = 65$$
$$x = 13$$

Ryan is 13 years old.
Kelsey is $13 - 6 = 7$ years old.

Their mother is $3 \times 13 = 39$ years old. Check: $13 + 7 + 39 = 59$

Presentation
Applying linear equations

Video
Equation problems

7.03

EXERCISE 7.03 ANSWERS ON P. 632

Equation problems U F PS R C

1 When 7 is subtracted from 4 times a certain number, the answer is 37. What is the number? PS R C

EXAMPLE 3

2 If 15 more than a number is 3 more than double the number, what is the number?

3 Two-thirds of a number is 16. What is the number?

4 When two-fifths of a number is increased by 15, the result is 27. What is the number?

5 A rectangle is 4 times as long as it is wide. The perimeter of the rectangle is 100 cm. Find the dimensions of the rectangle. PS R C

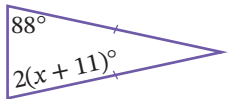
EXAMPLE 4

6 The length of a rectangle is 7 cm longer than its width.

a Let w be the width of the rectangle. Write an equation for w if the perimeter of the rectangle is 94 cm.

b Solve the equation and find the dimensions of the rectangle.

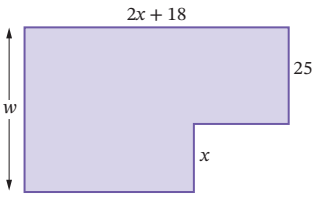
7 Find the value of x in this triangle.



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8 This block of land has a perimeter of 164 m.

Find the width (w) of the block. PS R C



9 Brad had to walk, run and then catch a bus to get to the train station 5.8 km away. The bus ride was 3.5 km and Brad walked 4 times as far as he ran. How far did Brad run in metres? PS R C

EXAMPLE 5

10 The sum of 2 consecutive numbers is 87. Find the numbers. PS R C

11 The sum of 3 consecutive numbers is 87. Find the numbers.

12 The sum of 3 consecutive even integers is 168. What are the 3 integers?

13 The sum of 3 consecutive *odd* integers is 75. Find the integers.

EXAMPLE 6

14 Dean's father, Franco, is 5 times Dean's age. Dean is 8 years older than his sister, Helen. The sum of all their ages is 62 years. How old is each person? (*Hint:* Let Dean's age be x.) PS R C

15 Bill is 3 times as old as his daughter, Rebekah, who is 6 years younger than her brother, Tyrone. How old is Rebekah if the sum of their 3 ages is 76 years?

16 In my money box I have only \$1 and \$2 coins. I have 240 coins in total, worth \$318. How many \$2 coins do I have in the money box?

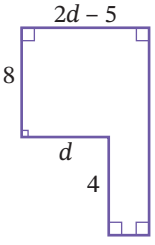
17 Aerin bought 4 ice creams and received \$2.80 change from his \$20 note. How much did each ice cream cost?

18 Ashvin is twice as tall as his little sister, Sureti, and 30 cm shorter than his father. Their combined height is 3.8 m. Find the height of each person.

19 The length of a rectangle is 9.5 cm longer than its width. The perimeter of the rectangle is 87 cm. Find the dimensions of the rectangle.

20 Janine is 6 years younger than her husband, Vila, who is 3 times the age of their son Brett. Brett is 5 years older than his sister Amanda. The sum of all their ages is 125 years. How old is each person?

21 A house is constructed in the shape shown below. If the house has a floor area of 54m^2, what is the value of d? PS R C



22 Ashleigh only collects 20-cent and 50-cent coins. Last week she counted 420 coins to the value of \$156. How many 50-cent coins did Ashleigh have?

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Mental skills 7A: Maths without calculators ANSWERS ON P. 632

Quiz
Mental skills 7A

Fraction of a quantity

Learn these commonly-used fractions and their decimal equivalents.

Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{2}{5}$
Decimal	0.5	0.25	0.125	0.75	0.2	0.1	0.05	0.4

Now we will use them to find a fraction or decimal of a quantity.

1 Study each example.

a $\frac{1}{4} \times 72 = 72 \div 4$
$= 18$

b $\frac{2}{5} \times 40 = (\frac{1}{5} \times 40) \times 2$
$= 8 \times 2$
$= 16$

c $\frac{3}{4} \times 32 = (\frac{1}{4} \times 32) \times 3$
$= 8 \times 3$
$= 24$

d $0.5 \times 66 = \frac{1}{2} \times 66$
$= 33$

e $0.05 \times 80 = \frac{1}{20} \times 80$
$= 4$

f $0.125 \times 56 = \frac{1}{8} \times 56$
$= 7$

2 Now simplify each expression.

a $\frac{1}{2} \times 28$ **b** $\frac{1}{4} \times 36$ **c** $\frac{1}{10} \times 70$ **d** $\frac{1}{8} \times 64$

e $\frac{1}{5} \times 15$ **f** $\frac{1}{10} \times 80$ **g** $\frac{2}{5} \times 25$ **h** $\frac{1}{20} \times 100$

i $\frac{3}{4} \times 44$ **j** $\frac{1}{8} \times 40$ **k** 0.25×60 **l** 0.4×45

m 0.1×260 **n** 0.125×48 **o** 0.75×48 **p** 0.05×120

q 0.2×70 **r** 0.5×320 **s** 0.25×56 **t** 0.125×16

Equations with algebraic fractions 7.04

Worksheet
Equations 3

Puzzle
Equations code puzzle

Video
Solving equations with fractions

Example 7

Solve each equation.

a $\frac{x+11}{4}+9=-3$

b $\frac{a-2}{4}=\frac{2a+5}{3}$

Solution

a $\frac{x+11}{4}+9=-3$

$\frac{x+11}{4}+9-9=-3-9$

$\frac{x+11}{4}=-12$

$\frac{x+11}{4}\times 4=-12\times 4$

$x+11=-48$

$x+11-11=-48-11$

$x=-59$

b $\frac{a-2}{4}=\frac{2a+5}{3}$

For equations where all terms are fractions, multiply both sides by a common multiple of the denominators to remove the fractions. The **lowest common multiple (LCM)** of 3 and 4 is 12, so multiply both sides by 12.

$12\times\frac{(a-2)}{4}=12\times\frac{(2a+5)}{3}$

$\cancel{12}^{3}\times\frac{(a-2)}{\cancel{4}_{1}}=\cancel{12}^{4}\times\frac{(2a+5)}{\cancel{3}_{1}}$

$3(a-2)=4(2a+5)$

$3a-6=8a+20$

$-5a-6=20$

$-5a=26$

$a=\frac{26}{-5}$

$a=-5\frac{1}{5}$

PATH

Example 8

Solve each equation.

a $\frac{k}{3}+\frac{k}{7}=4$

b $\frac{2m}{3}-\frac{m}{2}=2$

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Solution

a Multiply both sides by 21, the LCM of 3 and 7.

$$21\left(\frac{k}{3}+\frac{k}{7}\right)=21\times 4$$

$$21\times\frac{k}{3}+21\times\frac{k}{7}=84$$

$$^{7}\cancel{21}\times\frac{k}{\cancel{3}_{1}}+\cancel{21}^{3}\times\frac{k}{\cancel{7}_{1}}=84$$

$$7k+3k=84$$

$$10k=84$$

$$k=\frac{84}{10}$$

$$k=8\frac{2}{5}$$

Check by substituting that this solution is correct.

Note: We can also solve this equation by simplifying the LHS first:

$$\frac{k}{3}+\frac{k}{7}=4$$

$$\frac{7k}{21}+\frac{3k}{21}=4$$

$$\frac{10k}{21}=4$$

$$10k=84$$

$$k=\frac{84}{10}$$

$$k=8\frac{k}{3}$$

However, the first method may be faster and easier because operations with fractions are not needed.

b Multiply both sides by 6, the LCM of 3 and 2.

$$6\left(\frac{2m}{3}-\frac{m}{2}\right)=6\times 2$$

$$\cancel{6}^{2}\times\frac{2m}{\cancel{3}_{1}}-\cancel{6}^{3}\times\frac{m}{\cancel{2}_{1}}=12$$

$$4m-3m=12$$

$$m=12$$

PATH

Video Equations with algebraic fractions

7.04

Example 9

Solve each equation.

a $\frac{2n+1}{3}-\frac{3n-2}{2}=-5$

b $\frac{a+5}{6}+\frac{3a}{4}=5$

Solution

a Multiply both sides by 6, the LCM of 3 and 2.

$$6\left(\frac{2n+1}{3}-\frac{3n-2}{2}\right)=6\times(-5)$$

$$\cancel{6}^{2}\left(\frac{2n+1}{\cancel{3}_{1}}\right)-\cancel{6}^{3}\left(\frac{3n-2}{\cancel{2}_{1}}\right)=-30$$

$$2(2n+1)-3(3n-2)=-30$$

$$4n+2-9n+6=-30$$

$$-5n+8=-30$$

$$-5n=-38$$

$$n=\frac{-38}{-5}$$

$$=7\frac{3}{5}$$

b Multiply both sides by 12, the LCM of 6 and 4.

$$12\left(\frac{a+5}{6}+\frac{3a}{4}\right)=12\times 5$$

$$^{2}\cancel{12}\left(\frac{a+5}{\cancel{6}_{1}}\right)+^{3}\cancel{12}\left(\frac{3a}{\cancel{4}_{1}}\right)=60$$

$$2(a+5)+3(3a)=60$$

$$2a+10+9a=60$$

$$11a+10=60$$

$$11a=50$$

$$a=\frac{50}{11}$$

$$=4\frac{6}{11}$$

Video Equations with algebraic fractions

EXERCISE 7.04 ANSWERS ON P. 632

Equations with algebraic fractions U F R

EXAMPLE 7

1 Solve each equation. R

a $\frac{a+1}{4}+5=15$ **b** $\frac{x+4}{5}-10=2$ **c** $\frac{x-5}{5}-1=4$

d $\frac{y-2}{5}-2=6$ **e** $\frac{p+1}{3}-11=8$ **f** $5+\frac{4y+6}{9}=15$

g $1+\frac{10x-2}{2}=-10$ **h** $\frac{3y+5}{4}+9=-1$ **i** $\frac{12-x}{3}-1=6$

j $\frac{15-d}{7}+10=0$ **k** $\frac{3-2w}{11}+4=5$ **l** $\frac{6-4p}{9}-10=-6$

PATH

2 Solve each equation. R

a $\frac{x+5}{4}=\frac{x+4}{5}$ **b** $\frac{y+8}{3}=\frac{y-2}{2}$ **c** $\frac{2p-1}{4}=\frac{p+5}{3}$

d $\frac{2w+5}{3}=\frac{3w+1}{5}$ **e** $\frac{8x-4}{3}=\frac{4x+5}{2}$ **f** $\frac{m-6}{4}=\frac{2m-3}{10}$

g $\frac{2y+4}{3}=\frac{5y-2}{5}$ **h** $\frac{4x-2}{6}=\frac{x+7}{4}$ **i** $\frac{12-2x}{4}=\frac{9+x}{3}$

j $\frac{8-3w}{5}=\frac{2w+1}{4}$ **k** $\frac{7-5x}{2}=\frac{1-9x}{4}$ **l** $\frac{8-x}{2}=\frac{2x+1}{3}$

EXAMPLE 8

3 Solve $\frac{x}{3}+\frac{x}{2}=1$. Select the correct answer **A**, **B**, **C** or **D**. R

A $x=1\frac{1}{5}$ **B** $x=1\frac{1}{2}$ **C** $x=1$ **D** $x=2$

4 Solve each equation. R

a $\frac{d}{4}+\frac{d}{3}=14$ **b** $\frac{k}{2}-\frac{k}{3}=3$ **c** $\frac{m}{3}+\frac{m}{2}=10$

d $\frac{x}{8}+\frac{x}{3}=10$ **e** $\frac{p}{3}-\frac{p}{5}=4$ **f** $\frac{k}{4}-\frac{k}{5}=1$

g $\frac{4k}{3}-\frac{k}{5}=34$ **h** $\frac{3m}{5}+\frac{m}{2}=11$ **i** $\frac{4a}{5}+\frac{2a}{3}=10$

5 What is the solution to $\frac{p-1}{5}-\frac{p+3}{2}=4$? Select **A**, **B**, **C** or **D**. R

EXAMPLE 9

A $p=-12$ **B** $p=-19$ **C** $p=14$ **D** $p=3$

6 Solve each equation. R

a $\frac{x}{7}+\frac{x-1}{4}=0$ **b** $\frac{2p}{3}-\frac{p+1}{6}=2$

c $\frac{m+2}{2}+\frac{m+1}{3}=12$ **d** $\frac{c-2}{5}-\frac{c-3}{2}=4$

e $\frac{2y-1}{4}+\frac{y+1}{2}=6$ **f** $\frac{x-1}{3}+\frac{x-4}{6}=-5$

g $\frac{7+2p}{5}-\frac{1-p}{2}=-1$ **h** $\frac{6y-1}{4}-\frac{y+2}{3}=8$

i $\frac{a+5}{4}+3=\frac{2a}{5}$ **j** $\frac{3x-1}{4}=2x+\frac{5}{6}$

k $\frac{12u+3}{7}=2u-\frac{u+4}{6}$ **l** $\frac{m-6}{5}+2m=\frac{3m-1}{4}$

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Simple quadratic equations $ax^2 = c$ 7.05

An equation in which the highest power of the variable is 2 is called a **quadratic equation**, for example, $x^2 = 5$, $3m^2 + 7 = 10$, $d^2 - 4 = 0$ and $4t^2 - 3t = 8$.

Worksheet
Equations review

7.05

The quadratic equation $x^2 = c$

The quadratic equation $x^2 = c$ (where c is a positive number) has 2 solutions:

$$x = \pm\sqrt{c} \text{ (which means } x = \sqrt{c} \text{ and } x = -\sqrt{c}).$$

Video
Simple quadratic equations $ax^2 = c$

Example 10

Solve each quadratic equation.

a $y^2 = 16$ **b** $p^2 = 65$ **c** $5a^2 = 245$

Solution

a $y^2 = 16$

$y = \pm\sqrt{16}$

$= \pm 4$

b $p^2 = 65$

$p = \pm\sqrt{65}$

65 is not a square number, so leave the answer as a surd.

c $5a^2 = 245$

$a^2 = \frac{245}{5}$

$a^2 = 49$

$a = \pm\sqrt{49}$

$= \pm 7$

Example 11

Solve each quadratic equation, writing the solution correct to one decimal place.

a $4x^2 = 500$ **b** $\frac{3h^2}{5} = 26$

Solution

a $4x^2 = 500$

$x^2 = \frac{500}{4}$

$x^2 = 125$

$x = \pm\sqrt{125}$

$= \pm 11.1803...$

$\approx \pm 11.2$

b $\frac{3h^2}{5} = 26$

$3h^2 = 26 \times 5$

$3h^2 = 130$

$h^2 = \frac{130}{3}$

$h^2 = 43\frac{1}{3}$

$h = \pm\sqrt{43\frac{1}{3}}$

$= \pm 6.5828...$

$\approx \pm 6.6$

EXERCISE 7.05 ANSWERS ON P. 632

Simple quadratic equations $ax^2 = c$ U F R C

EXAMPLE 10

1 Solve each quadratic equation, writing the solutions as surds if necessary.

a $m^2 = 144$ **b** $x^2 = 400$ **c** $y^2 = 225$ **d** $k^2 = 59$

e $y^2 = 10$ **f** $w^2 = 36$ **g** $8x^2 = 200$ **h** $9t^2 = 81$

i $\frac{a^2}{2} = 8$ **j** $5k^2 = 40$ **k** $3w^2 = 30$ **l** $2d^2 = 288$

m $\frac{k^2}{2} = 8$ **n** $\frac{w^2}{10} = 7$ **o** $4x^2 = 1$ **p** $\frac{m^2}{3} = 27$

q $8y^2 = 40$ **r** $2p^2 + 3 = 21$ **s** $3k^2 = 48$ **t** $\frac{y^2}{5} - 2 = 9$

u $6x^2 = 42$

EXAMPLE 11

2 Solve each equation, writing the solution correct to one decimal place.

a $m^2 = 20$ **b** $b^2 = 17$ **c** $v^2 = 6$ **d** $2p^2 = 35$

e $9k^2 = 63$ **f** $\frac{x^2}{5} = 8$ **g** $\frac{k^2}{16} = 6$ **h** $\frac{7u^2}{10} = 2$

i $6y^2 = 84$ **j** $\frac{3w^2}{4} = 20$ **k** $a^2 + 11 = 28$ **l** $2y^2 - 14 = 63$

3 Explain why the quadratic equation $k^2 + 25 = 0$ has no solutions. R C

4 State which of these quadratic equations has no solutions. Give reasons. R C

a $x^2 = -9$ **b** $2k^2 + 5 = 9$ **c** $3m^2 + 8 = 4$

d $\frac{9w^2}{2} - 1 = 1$ **e** $4 + \frac{d^2}{3} = 8$ **f** $\frac{5a^2}{2} + 3 = 2$

7.06 Quadratic equations $x^2 + bx + c = 0$

PATH

Video Quadratic equations by factorising

The quadratic equation $x^2 + bx + c = 0$

To solve quadratic equations of the form $x^2 + bx + c = 0$, factorise $x^2 + bx + c$ into a binomial product of the form $(x + d)(x + e)$.

Example 12

Solve $x^2 + 7x + 12 = 0$

Solution

To factorise $x^2 + 7x + 12$, find 2 numbers that have a sum of 7 and a product of 12.
The correct numbers are 3 and 4.
$\therefore x^2 + 7x + 12 = 0$
$(x + 3)(x + 4) = 0$

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The LHS has been factorised into 2 binomials, $(x + 3)$ and $(x + 4)$, whose product is 0.
If 2 numbers have a product of 0, then one of the numbers *must* be 0.
$\therefore x + 3 = 0$ or $x + 4 = 0$
$\therefore x = -3$ or $x = -4$
$\therefore$ The solution to $x^2 + 7x + 12 = 0$ is
$x = -3$ or $x = -4$

Check:
When $x = -3$,
LHS $= (-3)^2 + 7 \times (-3) + 12 = 0$
RHS $= 0$
LHS = RHS

When $x = -4$,
LHS $= (-4)^2 + 7 \times (-4) + 12 = 0$
RHS $= 0$
LHS = RHS

Example 13

Solve each quadratic equation.

a $m^2 - 5m - 14 = 0$

b $d^2 + 3d - 40 = 0$

c $w^2 = 15w$

d $k^2 - 12k + 36 = 0$

Solution

a $m^2 - 5m - 14 = 0$

Find 2 numbers that have a sum of -5 and a product of -14.
They are -7 and 2.
$(m - 7)(m + 2) = 0$
$\therefore m - 7 = 0$ or $m + 2 = 0$
$\therefore m = 7$ or $m = -2$
$\therefore$ The solution to $m^2 - 5m - 14 = 0$ is $m = 7$ or $m = -2$.
Check:
When $m = 7$,
LHS $= (7)^2 - 5 \times (7) - 14 = 0$
RHS $= 0$
LHS = RHS

When $m = -2$,
LHS $= (-2)^2 - 5 \times (-2) - 14 = 0$
RHS $= 0$
LHS = RHS

b $d^2 + 3d - 40 = 0$
$(d - 5)(d + 8) = 0$
$\therefore d - 5 = 0$ or $d + 8 = 0$
$\therefore d = 5$ or $d = -8$
$\therefore$ The solution to $d^2 + 5d - 40 = 0$ is $d = 5$ or $d = -8$.

PATH

c $w^2 = 15w$

$w^2 - 15w = 0$

You cannot divide both sides by w because if $w = 0$ then you are dividing by 0.

This requires a simpler factorisation as there are only 2 terms, both involving w.

$w(w - 15) = 0$

$\therefore w = 0$ or $w - 15 = 0$

$\therefore w = 0$ or $w = 15$

$\therefore$ The solution to $w^2 - 15w = 0$ is $w = 0$ or $w = 15$.

d $k^2 - 12k + 36 = 0$

$(k - 6)(k - 6) = 0$

$\therefore k - 6 = 0$ or $k - 6 = 0$

$\therefore k = 6$ or $k = 6$

$\therefore$ The solution to $k^2 - 12k + 36 = 0$ is $k = 6$ (only one solution).

Note: Quadratic equations of the form $ax^2 + bx + c = 0$ will be met in Chapter 10, *Quadratic equations and the parabola*, of *New Century Maths 10 Advanced*.

EXERCISE 7.06 ANSWERS ON P. 632

Quadratic equations $x^2 + bx + c = 0$ U F

EXAMPLE 12

1 Solve each equation.

a $h^2 + 5h + 4 = 0$ **b** $y^2 + 9y + 20 = 0$ **c** $g^2 + 10g + 24 = 0$

d $p^2 + 15p + 56 = 0$ **e** $m^2 + 19m + 88 = 0$ **f** $u^2 + 12u + 32 = 0$

EXAMPLE 13

2 Solve each equation.

a $x^2 - 3x - 4 = 0$ **b** $n^2 + 8n - 33 = 0$ **c** $y^2 + 4y - 96 = 0$

d $w^2 - 9w - 10 = 0$ **e** $a^2 + 14a - 51 = 0$ **f** $v^2 - 2v - 35 = 0$

3 Solve each quadratic equation. Select the correct answer **A**, **B**, **C** or **D**.

a $x^2 - 8x = 0$

A $x = 4, -4$ **B** $x = 0, -4$ **C** $x = 0, -8$ **D** $x = 8, 0$

b $q^2 + 5q = 0$

A $q = 5, -5$ **B** $q = 10, -5$ **C** $q = 0, -5$ **D** $q = 5, 0$

c $h^2 = 21h$

A $h = 21, 0$ **B** $h = 3, -7$ **C** $h = 7, -3$ **D** $h = 0, -21$

4 Solve each equation.

a $g^2 - 4g + 4 = 0$ **b** $m^2 + m = 0$ **c** $k^2 = 12k$

d $p^2 - 14p + 49 = 0$ **e** $a^2 + 8a + 16 = 0$ **f** $y^2 = 24y$

5 Solve each equation.

a $d^2 - 9d - 36 = 0$ **b** $y^2 + 16y + 63 = 0$ **c** $m^2 = 2m$

d $w^2 + 8w + 15 = 0$ **e** $a^2 - 4a = 0$ **f** $p^2 - 11p + 30 = 0$

g $e^2 - 24e + 144 = 0$ **h** $c^2 - 8c - 20 = 0$ **i** $s^2 + 10s + 25 = 0$

j $b^2 = 9b$ **k** $u^2 + 13u + 40 = 0$ **l** $g^2 + 6g - 7 = 0$

m $r^2 - 2r + 1 = 0$ **n** $w^2 + 8w = 0$ **o** $x^2 - x - 72 = 0$

6 Solve each equation.

a $6x^2 = 42x$ **b** $3p^2 + 15p + 18 = 0$ **c** $2k^2 - 4k - 48 = 0$

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Investigation

PATH

Solving $x^3 = c$

1 $x^3 = 27$ has only one solution. What is it?

2 What is the solution to each equation?

a $x^3 = 125$ **b** $x^3 = 64$ **c** $x^3 = -8$

3 What is the inverse operation of 'cubing'?

4 Study this example:

$x^3 = -27$

$x = \sqrt[3]{-27}$

$x = -3$

The cube root of a negative number is also negative.

Check: When $x = -3$, $x^3 = (-3)^3 = -27$

Now use the same method to solve each equation and check your answers.

a $r^3 = -1000$ **b** $u^3 = 216$

5 How many solutions does each cubic equation have?

a $n^3 = 1$ **b** $t^3 = -343$ **c** $x^3 = -1$

7.07

Simple cubic equations $ax^3 = c$

PATH

An equation in which the highest power of the variable is 3 is called a **cubic equation**, for example, $x^3 = 12$, $2m^3 + 1 = 25$, $d^3 - 14 = 4$ and $x^3 - 3x^2 + 5x + 4 = 0$.

The cubic equation $x^3 = c$

The cubic equation $x^3 = c$ has one solution:

$$x = \sqrt[3]{c}$$

PATH

Example 14

Solve each cubic equation.

a $n^3 = 729$ **b** $d^3 = 40$ **c** $-3y^3 = 1029$

Solution

a
$n^3 = 729$ Finding the cube root of both sides
$n = \sqrt[3]{729}$
$= 9$

b
$d^3 = 40$ Finding the cube root of both sides.
$d = \sqrt[3]{40}$ 40 is not a cube number so leave the answer as a surd.

c
$-3y^3 = 1029$ Dividing both sides by -3
$y^3 = \frac{1029}{-3}$
$y^3 = -343$
$y = \sqrt[3]{-343}$
$= -7$

Example 15

Solve each cubic equation, writing the solution correct to one decimal place.

a $2h^3 = -18$ **b** $\frac{z^3}{7} = 11$

Solution

a
$2h^3 = -18$
$h^3 = \frac{-18}{2}$
$h^3 = -9$
$h = \sqrt[3]{-9}$
$= -2.0800\ldots$
≈ -2.1

b
$\frac{z^3}{7} = 11$
$z^3 = 11 \times 7$
$z^3 = 77$
$z = \sqrt[3]{77}$
$= 4.2543\ldots$
≈ 4.3

EXERCISE 7.07 ANSWERS ON P. 632

Simple cubic equations $ax^3 = c$ UFRC

EXAMPLE 14

1 Solve each cubic equation, writing the solutions as surds if necessary.

a $r^3 = 1$ **b** $k^3 = 216$ **c** $d^3 = -27$

d $x^3 = -45$ **e** $w^3 = 100$ **f** $f^3 = 64$

g $4m^3 = 32$ **h** $2t^3 = -250$ **i** $\frac{c^3}{2} = 108$

j $-3k^3 = 192$ **k** $7a^3 = 105$ **l** $\frac{2p^3}{9} = -4$

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m $4q^3 = 665.5$ **n** $\frac{n^3}{5} = -345.6$ **o** $8s^3 = 150$

p $\frac{e^3}{8} = -1$ **q** $-6y^3 = 40$ **r** $2v^3 - 10 = 1014$

s $\frac{4b^3}{5} = 12$ **t** $\frac{z^3}{6} + 4 = -8$ **u** $-4x^3 = -144$

2 Solve each equation, writing the solution correct to one decimal place.

a $c^3 = 47$ **b** $g^3 = 151$ **c** $y^3 = -450$

d $-8p^3 = 728$ **e** $3u^3 = 245$ **f** $\frac{x^3}{5} = -11$

g $-\frac{h^3}{20} = 25$ **h** $\frac{11a^3}{7} = 80$ **i** $-6d^3 = 186$

j $\frac{5v^3}{8} = -27$ **k** $a^3 - 45 = 220$ **l** $4j^3 + 72 = -166$

3 **a** Does a cubic equation of the form $x^3 = c$ always have a solution?

b When is the solution to $x^3 = c$ positive?

c When does $x^3 = c$ have 2 solutions? R C

PATH

EXAMPLE 15

7.08

Equations and formulas 7.08

A **formula** is an algebraic equation that shows a relationship between variables. For example, the formula for the area of a circle is $A = \pi r^2$, where A is the area and r is the radius of the circle (π is a constant). Because the formula is for the area, A is called the **subject** of the formula and it is the variable on its own on the left side of the '=' sign.

Worksheet Working with formulas

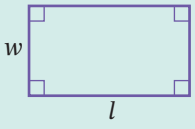
Videos Equations and formulas 1

Equations and formulas 2

Example 16

The formula for the perimeter (P) of a rectangle of length l and width w is given by

$$P = 2(l + w).$$



Use the formula to find:

a the perimeter of a rectangle with length 20 cm and width 9 cm

b the width of a rectangle if its length is 12 m and its perimeter is 70 m

c the length of a rectangle if its width is 42 cm and its perimeter is 1.8 m.

Solution

a $l = 20, w = 9$:

$P = 2(l + w)$

$= 2(20 + 9)$

$= 2 \times 29$

$= 58$

$\therefore$ The perimeter is 58 cm.

b $l = 12, P = 70$:

$P = 2(l + w)$

$70 = 2(12 + w)$

$70 = 24 + 2w$

$46 = 2w$

$w = \frac{46}{2}$

$= 23$

$\therefore$ The width is 23 cm.

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c $w = 42, P = 1.8 \text{ m} = 180 \text{ cm}$ since w is given in cm

$P = 2(l + w)$

$180 = 2(l + 42)$

$180 = 2l + 84$

$96 = 2l$

$l = \frac{96}{2}$

$= 48$

∴ The length is 48 cm.

Example 17

The cost of hiring a sound system for a party is $250 plus $75 per hour. The cost can be represented by the formula $C = 250 + 75h$, where C is the total cost (in dollars), and h is the number of hours.

a Find the cost of hiring the sound system for 4 hours.

b A family is willing to spend $750 for hiring the sound system. What is the maximum number of *whole* rental hours that the family can afford?



iStock.com/Bet_Noire

Solution

a $h = 4$:

$C = 250 + 75h$

$= 250 + 75 \times 4$

$= 550$

∴ The cost is $550.

b $C = 750$:

$750 = 250 + 75h$

$500 = 75h$

$h = \frac{500}{75}$

$= 6\frac{2}{3}$

∴ The maximum number of whole hours is 6.

EXERCISE 7.08 ANSWERS ON P. 633

Equations and formulas U F PS R

1 Given the formula $y = 5x + c$, find:

a y if $x = 5$ and $c = 3$

b y if $x = -1$ and $c = 16$

c c if $y = 40$ and $x = 3$

d c if $y = -6$ and $x = -1$

e x if $y = 27$ and $c = 12$

f x if $y = 64$ and $c = -16$

EXAMPLE 16

2 A temperature in degrees Celsius (°C) can be converted to degrees Fahrenheit (°F) using the formula $F = \frac{9C}{5} + 32$. Convert each temperature to °F.

a 35°C **b** −10°C **c** 16°C

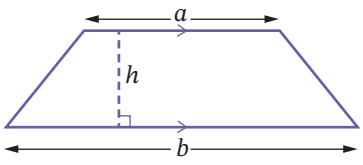
Foundation Standard Complex

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3 Use the formula in question **2** to convert each temperature to °C, correct to one decimal place. PS R

a 100°F **b** −45°F **c** 78°F

4 The formula $A = \frac{1}{2}(a + b)h$ is used to find the area of a trapezium, where A is the area, a and b are the lengths of the parallel sides, and h is the perpendicular height between them.



Use the formula to find:

a the area of a trapezium with height 6 cm and parallel sides of length 9 cm and 15 cm

b the height of a trapezium if its area is 420 cm^2 and it has parallel sides of length 22 cm and 20 cm

c the length of one side of a trapezium if its parallel side is 20.5 m, its area is 318 m^2 and its height is 12 m.

5 Find the value of S in the formula $M = \frac{kS}{5}$ if $M = 12.6$ and $k = 3.15$. Select **A**, **B**, **C** or **D**.

A 60 **B** 7.938 **C** 20 **D** 0.8

6 The cost C (in dollars) of hiring a limousine is given by the formula $C = 2500 + 150h$, where h is the number of hours of hire. Find: PS R

a the cost of hiring a limousine for 4 hours

b the cost of hiring a limousine for 2 days

c the number of hours for which you could hire a limousine for $4000

d the maximum number of whole hours for which a limousine could be hired at a cost of $5250

7 The profit, $P, made by an online store selling games is given by $P = 35x - 750$, where x represents the number of games sold. Find: PS R

a the profit made when 230 games are sold

b the number of games sold if the profit is $18 010.

8 A catering company charges $C for a function with P people using the formula $C = 1500 + 75.5P$. PS R

a How much does the company charge for a function with 10 guests?

b Find the cost of catering for a group of 60 people.

c Diane has $10 000 to spend on catering for her wedding reception. What is the maximum number of people she can invite?

9 The perimeter of a rectangle is given by the formula $P = 2(l + w)$, where l is the rectangle's length and w is its width. What is the length of a rectangle with a width of 4 and a perimeter of 52? Select **A**, **B**, **C** or **D**.

A 9 **B** 12 **C** 18 **D** 22

7.08

EXAMPLE 17

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10 The temperature T (in °C) of a hot liquid as it cools is given by the formula $T = 100 - 17.5h$, where h is the number of hours it has been cooling. Find: PS R

a the temperature of the liquid after 2 hours

b the temperature of the liquid after 30 minutes

c the number of hours it takes for the temperature of the liquid to reach 30°C.

11 Archeologists use the formula $H = 2.52t + 75.8$ to estimate the height, H cm, of a man when the tibia (shin bone) length, t cm, is measurable. PS R

a A male tibia 42 cm long was found intact. Estimate the height of the male, correct to the nearest centimetre.

b Estimate, to the nearest centimetre, the length of the tibia of a male whose height is 174 cm.

Quiz
Mental skills 7B

Mental skills 7B: Maths without calculators ANSWERS ON P. 633

Percentage of a quantity

Learn these commonly-used percentages and their fraction equivalents.

Percentage	50%	25%	12.5%	75%	20%	10%	$33\frac{1}{3}\%$	$66\frac{2}{3}\%$
Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{3}$	$\frac{2}{3}$

Now we will use them to find a percentage of a quantity.

1 Study each example.

a $20\% \times 25 = \frac{1}{5} \times 25$
$= 5$

b $50\% \times 120 = \frac{1}{2} \times 120$
$= 60$

c $12.5\% \times 32 = \frac{1}{8} \times 32$
$= 4$

d $75\% \times 56 = \frac{3}{4} \times 60$
$= (\frac{1}{4} \times 60) \times 3$
$= 15 \times 3$
$= 45$

e $33\frac{1}{3}\% \times 27 = \frac{1}{3} \times 27$
$= 9$

f $66\frac{2}{3}\% \times 60 = \frac{2}{3} \times 60$
$= (\frac{1}{3} \times 60) \times 2$
$= 20 \times 2$
$= 40$

2 Now simplify each expression.

a $25\% \times 44$	**b** $33\frac{1}{3}\% \times 120$	**c** $20\% \times 35$	**d** $66\frac{2}{3}\% \times 36$
e $10\% \times 230$	**f** $12\frac{1}{2}\% \times 48$	**g** $50\% \times 86$	**h** $20\% \times 400$
i $75\% \times 24$	**j** $33\frac{1}{3}\% \times 45$	**k** $25\% \times 160$	**l** $10\% \times 650$
m $12.5\% \times 88$	**n** $66\frac{2}{3}\% \times 21$	**o** $20\% \times 60$	**p** $75\% \times 180$

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Changing the subject of a formula 7.09

In the formula $v = u + at$, v is the **subject** of the formula. When the formula is rearranged so that one of the other variables becomes the subject, the process is called **changing the subject of the formula**. To change the subject of a formula, use the same rules as for solving an equation. The answer is not a number but an algebraic expression (another formula).

Example 18

a For the formula $v = u + at$, change the subject to a

b Make b the subject of the formula $x = \sqrt{b^2 - 4ac}$.

c Given $P = \dfrac{m-1}{m+1}$, change the subject of the formula to m.

Solution

a To make a the subject of the formula, solve it like an equation for a.

$v = u + at$

$u + at = v$	Swapping sides so the new subject a appears on the LHS
$at = v - u$	Subtracting u from both sides
$a = \dfrac{v-u}{t}$	Dividing both sides by t

b $x = \sqrt{b^2 - 4ac}$

$\sqrt{b^2 - 4ac} = x$	Swapping sides
$b^2 - 4ac = x^2$	Squaring both sides
$b^2 = x^2 + 4ac$	Adding $4ac$ to both sides
$b = \pm\sqrt{x^2 + 4ac}$	Taking the positive and negative square root of both sides

c $P = \dfrac{m-1}{m+1}$

$\dfrac{m-1}{m+1} = P$	Swapping sides
$m - 1 = P(m+1)$	Multiplying both sides by $m + 1$
$= Pm + P$	Expanding
$m - Pm = P + 1$	Moving the m-terms to the LHS, the 1 to the RHS
$m(1 - P) = P + 1$	Factorise m from the LHS
$m = \dfrac{P+1}{1-P}$	Dividing both sides by $1 - P$

PATH

Puzzle Formulas squaresaw

Worksheet Changing the subject of a formula

Video Changing the subject of a formula

PATH

EXERCISE 7.09 ANSWERS ON P. 633

Changing the subject of the formula U F R C

EXAMPLE 18

1 Change the subject of each formula to x. R C

a $d = x + c$ **b** $y = mx + c$ **c** $p = ax - y$

d $y = \frac{x}{m} + p$ **e** $k = mrx$ **f** $v = \frac{m}{x}$

g $A = \frac{x+y}{2}$ **h** $c = ax + by$ **i** $A = \frac{1}{2}h(x+y)$

2 The volume of a pyramid has the formula $V = \frac{1}{3}Ah$, where A is the area of the base and h is the perpendicular height. Change the subject of the formula to A.

Select the correct answer **A**, **B**, **C** or **D**. R C

A $A = \frac{1}{3}Vh$ **B** $A = \frac{3}{Vh}$ **C** $A = 3Vh$ **D** $A = \frac{3V}{h}$

3 Make y the subject of each formula. R C

a $m = an + ay$ **b** $x = y^2$ **c** $k = \frac{x}{y}$

d $Q = P + \sqrt{y}$ **e** $x^2 + y^2 = r^2$ **f** $M = ny^2$

g $m = \frac{r}{\sqrt{y}}$ **h** $b = c^2 - 2ay$ **i** $t = \sqrt{y - x}$

4 Change the subject of each formula to the variable shown. R C

a $V = lbh;\ h = ?$ **b** $A = \pi r^2;\ r = ?$ **c** $K = \frac{1}{2}mv^2;\ v = ?$

d $F = \frac{9C}{5} + 32;\ C = ?$ **e** $s = ut + \frac{1}{2}at^2;\ a = ?$ **f** $A = \frac{h}{2}(x+y);\ y = ?$

5 What is the correct formula for $p = m(n + x)$ if x is the subject?

Select **A**, **B**, **C** or **D**. R C

A $x = p - mn$ **B** $x = \frac{p}{mn}$ **C** $x = \frac{p}{m} - n$ **D** $x = \frac{p}{n} - m$

6 The cost ($\$C$) of a hire car is given by $C = 80 + 4.2d$, where d is the distance travelled in kilometres. R C

a Find the cost of hiring the car for a journey of 50 km.

b Make d the subject of the formula.

c Find the number of whole kilometres that can be travelled in the hire car for $402.

7 The angle sum of a shape with n sides is $A°$, where $A = 180(n - 2)$. R C

a Use the formula to find the angle sum of a shape with 7 sides.

b Make n the subject of the formula.

c If the angle sum of a polygon is 1440°, how many sides does it have?

PATH

8 The body-mass index of an adult is given by the formula $B = \frac{m}{h^2}$, where m is the adult's mass in kilograms and h is their height in metres. R C

a Change the subject of the formula to h.

b Hence find, correct to 2 decimal places, the height of a person with a body mass index of 25 and a mass of 60 kg.

9 Make r the subject of the formula $\frac{1}{x} = \frac{1}{r} + \frac{1}{s}$. R C

10 Solve each equation for a. R C

a $\frac{a}{5} = ap + q$ **b** $ar = 3(a + b)$ **c** $Da = D - 2a$

d $p = \frac{ax}{a + y}$ **e** $k = \frac{1 - a}{1 + a}$ **f** $M(a + b) = N(a - b)$

Power plus ANSWERS ON P. 633

1 Solve each equation.

a $\frac{5}{2y} = 14$ **b** $\frac{7}{r} = \frac{1}{r - 3}$ **c** $\frac{y^2 + 8}{12} = 4\frac{2}{3}$

d $4m^2 - 3 = m^2 + 72$ **e** $\frac{3(m + 5)}{4} - \frac{2(m - 1)}{3} = m - 10 + \frac{m - 6}{2}$

2 Given $W = \frac{X - Y}{X + Y}$, find:

a W when $X = 15$ and $Y = 10$ **b** W when $X = 6$ and $Y = -12$

c X when $W = 25$ and $Y = 6$ **d** Y when $W = 5$ and $X = 1$

3 Karlie is a farmer who raises pigs and chickens. From a total of 42 animals, she can count 116 legs. Write an equation and solve it to find how many chickens she has.

4 Joe is twice as old as his daughter, Wendi. 10 years ago, he was 3 times as old as her. Write an equation and solve it to calculate how old Wendi is now.

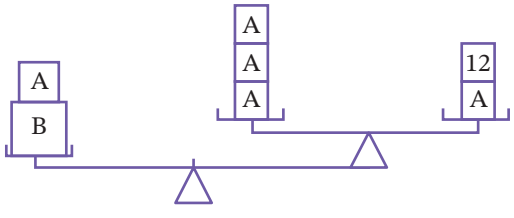
5 Consider $x^2 + y^2 = 4$.

a Explain why the smallest value for x is -2 and the largest value for x is 2.

b Are there any restrictions on the values that y can take? Explain why.

c By making y the subject, show that $y = \pm\sqrt{4 - x^2}$.

6 Find the value of B in the balance scale below.



7 Solve each cubic equation.

a $x^2 = x$ **b** $x^2 = x + 6$ **c** $x^3 = x$

d $x^3 = -x^2$ **e** $x^3 = 6x^2 - 9x$

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CHAPTER 7 REVIEW

Quiz
Language of maths 7

Puzzle
Equations crossword

Language of maths

backtracking method	balancing method	brackets
consecutive	cube root	equation
expand	flowchart	formula
inverse operation	LHS	lowest common multiple (LCM)
linear equation	quadratic equation	RHS
solution	solve	square root
subject	substitute	surd
undoing	unknown	variable

1 Which method of solving equations involves using inverse operations on both sides of the equation?

2 What is the **subject** of the formula $A = \frac{1}{2}(a + b)h$?

3 What name is given to numbers that follow each other in order, such as 9, 10, 11?

4 Write an example of:

a a quadratic equation **b** a linear equation

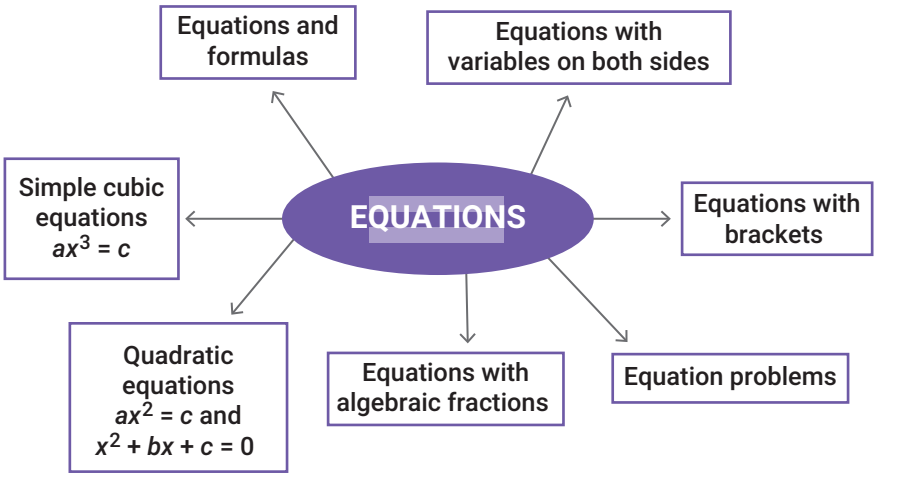
5 How many solutions does a linear equation have?

6 What does **LHS** stand for?

Topic summary

Worksheet
Mind map: Equations (Advanced)

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 7

ANSWERS ON P. 633

Quiz
Test yourself 7

1 Solve each equation. Check your solutions. 7.01

a $3x + 4 = 2x + 7$ **b** $5n - 3 = 2n - 15$ **c** $2d - 8 = -5d - 71$

d $4t = 12 - 4t$ **e** $8j - 17 = 10j$ **f** $6 - 3q = 8 - q$

2 Write an equation with x on both sides that has the solution $x = 3$. 7.01

3 Solve each equation. 7.02

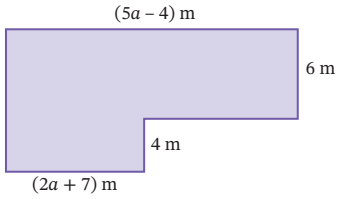
a $2(w - 5) = 4$ **b** $3(1 + 4n) = 15$ **c** $5(1 - 3p) = 20$

d $2(3 + x) = 5(x + 1)$ **e** $3(1 - y) = 4(2 - y)$ **f** $2(3 - 4x) = -(2x + 3)$

g $18 - 3a = 5a - 2(a - 6)$ **h** $11g + 2(g + 3) = 5(2g + 10)$

4 The length of a rectangle is 6 cm longer than it is wide. The perimeter of the rectangle is 76 cm. Find the dimensions of the rectangle. 7.03

5 This diagram shows the floor plan of a house. 7.03
If the area of the floor is 99 m^2, find the value of a.



6 If 6 more than a number equals 5 more than double the number, what is the number? 7.03

PATH

7 Solve each equation. 7.04

a $\frac{d-4}{2} + 9 = 4$ **b** $\frac{8n+6}{3} = \frac{6n+5}{2}$ **c** $\frac{2p}{3} - \frac{p}{2} = 4$

8 Solve each quadratic equation. 7.05

a $d^2 = 64$ **b** $8p^2 = 288$ **c** $3z^2 = 105$

PATH

9 Solve each quadratic equation. 7.06

a $w^2 - 6w + 5 = 0$ **b** $m^2 = 16m$ **c** $u^2 - 3u - 18 = 0$

10 Solve each cubic equation. 7.07

a $x^3 = 1331$ **b** $-4h^3 = 864$ **c** $\frac{t^3}{2} = 62.5$

11 The body mass index (BMI) of an adult is $B = \frac{M}{h^2}$, where M is the mass in kilograms and h is the height in metres. 7.08

a Find the BMI of Ned who is 1.85 m tall and has a mass of 72 kg.

b Find the mass of a person with a BMI of 24, who is 2.1 m tall.

12 The cost, C, in dollars, of hiring a taxi is $C = 5 + 2.4d$, where d is the distance travelled in kilometres. Find: 7.08

a the cost of a taxi trip if the distance travelled is 15 km

b the distance travelled if the cost of a taxi trip was \$78.20.

PATH

13 Make w the subject of each formula. 7.09

a $a = kw + v$ **b** $p = m(t - w)$ **c** $x = \frac{1}{w} + y$

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PRACTICE SET 2

ANSWERS ON P. 633

5.09 **1** State the number of significant figures in each number.

a 1203 **b** 0.008 **c** 1.092 **d** 15 700

5.01 **2** Simplify each expression.

a $y^9 \times y$ **b** $3g^7 \times 4g^8$ **c** $2p^3q \times 9p^2q^2$

d $\frac{k^{18}}{k^6}$ **e** $\frac{x^{11}}{x^7}$ **f** $h^{10} \div h$

g $54k^7 \div 6k^3$ **h** $\frac{h^3y^7}{hy^4}$ **i** $12m^5n^8 \div 18mn^7$

5.04 **3** Simplify each expression.

a $6y^0$ **b** $4^0 \times 3^0$ **c** $(4k)^0$ **d** $x \times (3y)^0$ **e** $2q^0 - (2q)^0$

4.09 **4** Sketch on a compass rose point A if it has a bearing of:

a 260° from point M **b** N56°W from point W.

5.05 **5** Evaluate each expression, giving answers in fraction form.

a 2^{-1} **b** 5^{-3} **c** 10^{-2}

6.03 **6** **a** Calculate the size of one angle in a regular hexagon.

b Name the polygon that has an angle sum of 1440°.

PATH

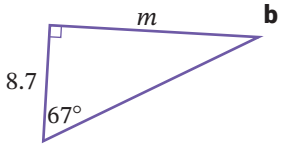
5.05 **7** Simplify each expression.

a $\left(\frac{1}{2}\right)^{-1}$ **b** $\left(\frac{8}{3}\right)^{-2}$ **c** $\left(\frac{4d}{5}\right)^{-2}$ **d** $\left(\frac{3}{10x}\right)^{-3}$

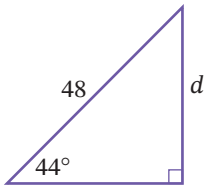
7.09 **8** Make y the subject of the formula $T = a + py^2$.

4.05 **9** Calculate the value of each variable, correct to 2 decimal places.

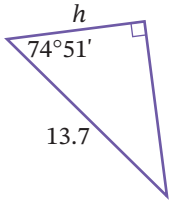
a



b



c



5.02 **10** Simplify each expression, giving the answers in index notation.

a $(2^3)^5$ **b** $(4^2)^6$ **c** $(3p^7)^2$ **d** $(2w^5)^3$

5.10 **11** Express each number in decimal form.

a 1.54×10^7 **b** 3.2×10^{-3} **c** 9.837×10^{-1} **d** 2.4×10^4

4.10 **12** A plane flies on a bearing of S85°W for 300 km. How far south (correct to the nearest kilometre) from its starting point has the plane flown?

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13 Find θ, correct to the nearest minute if: 4.07

a $\cos\theta = 0.21$ **b** $\sin\theta = \frac{9}{10}$ **c** $\tan\theta = 1.5$

14 Write each expression using a negative index. 5.05

a $\frac{1}{n}$ **b** $\frac{1}{2^3}$ **c** $\frac{1}{k^5}$ **d** $\frac{7}{y^2}$

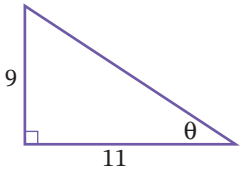
15 Write 7 280 000 in scientific notation. 5.10

16 Evaluate each expression, giving your answers in scientific notation correct to 2 significant figures. 5.11

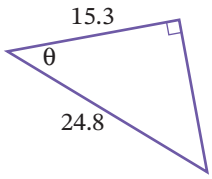
a $(7.08 \times 10^7) \times (3.8 \times 10^8)$ **b** $(4.65 \times 10^{-3}) \div (3.9 \times 10^7)$

17 Find θ, correct to the nearest degree. 4.07

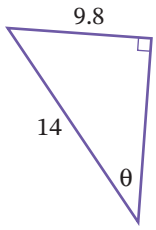
a



b

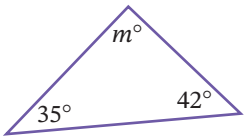


c

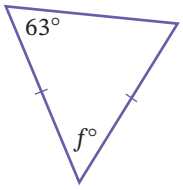


18 Find the value of each variable, giving reasons. 6.01

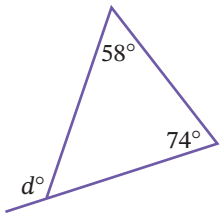
a



b



c



19 Solve $12 - 5k = 2k - 9$. Select the correct answer **A**, **B**, **C** or **D**. 7.01

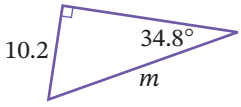
A $k = 3$ **B** $k = 1$ **C** $k = \frac{3}{7}$ **D** $k = 7$

20 Select the larger number for each pair of numbers. 5.10

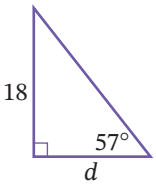
a 5.3×10^9 or 8.6×10^8 **b** 3.73×10^5 or 3.4×10^5

21 Find the value of each variable, correct to one decimal place. 4.06

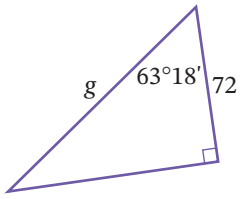
a



b



c

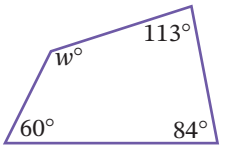


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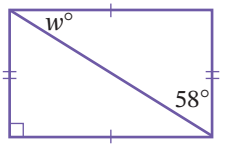
6.02

22 Find the value of w in each quadrilateral.

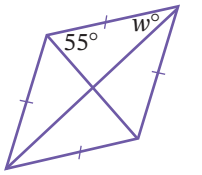
a



b



c



PATH

5.07

23 Write each expression using a fractional index.

a $\sqrt{13}$ **b** $\sqrt[3]{20}$ **c** $\sqrt[5]{3p}$ **d** $\left(\sqrt{x}\right)^3$

7.08

24 The volume of useable timber (in cubic metres) in a tree is calculated using the formula:

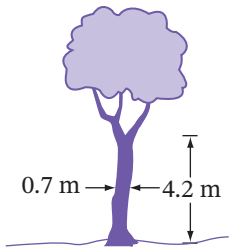
$$v = 0.5hd^2 + 10$$

where: d = the diameter of the tree

h = the height to the first branch

Calculate the volume of useable timber in this tree.

Select **A, B, C** or **D.**



A 16.2 m^3 **B** 11.5 m^3 **C** 11.0 m^3 **D** 18.8 m^3

PATH

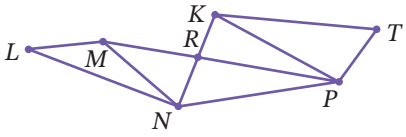
5.06

25 Evaluate each expression.

a $25^{\frac{3}{2}}$ **b** $4^{-\frac{5}{2}}$ **c** $\left(\sqrt[4]{16}\right)^3$ **d** $\sqrt[3]{27^2}$

6.08

26 Verify Euler's formula for this planar graph.



6.07

27 This graph shows a road network between 8 towns.

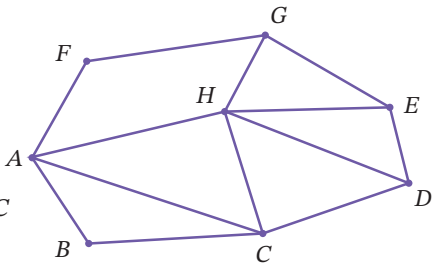
a Find the degree of each vertex.

b State whether each walk is a trail or path

i *ACHGEHA* **ii** *HDEGFA*

iii *AHDCBA* **iv** *HDEGFAHC*

c Why is the graph a planar graph?



7.04

28 Solve each equation.

a $\frac{3k-1}{2} = \frac{5k+1}{3}$ **b** $\frac{h}{5} + \frac{h}{2} = 6$

5.09

29 Round:

a 43 759 to 2 significant figures

b 10.7614 to 3 significant figures

c 0.000 038 9 to one significant figure

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30 In $\triangle ABC$, $\angle A = 90°$, $AC = 12.7$ cm and $BC = 17.8$ cm. Find $\angle B$, correct to the nearest minute. 4.07

31 A formula for cooking roast beef is $T = 40M + 25$, where T is the cooking time in minutes and M is the mass of the beef in kilograms. 7.08

a How long will it take to cook a 5 kg piece of beef?

b What mass of beef has a cooking time of $2\frac{1}{2}$ hours?

32 A 3.5 m ladder leans against a wall, making an angle of 50° with the ground. 4.05

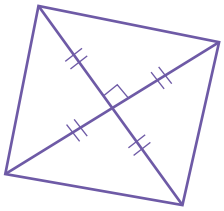
a Draw a diagram showing this information.

b Calculate, correct to the nearest 0.1 m:

i how far up the wall the ladder reaches

ii how far the foot of the ladder is from the base of the wall

33 What type of quadrilateral is shown here? 6.02



PATH

34 Find the new position of the point M (−5, −2) if it is: 6.06

a reflected in the y-axis

b rotated clockwise through 90° about the origin

c translated 7 units right and 3 units down

d reflected in the x-axis and then rotated 180° about the origin.

35 **a** The sum of 4 consecutive numbers is 82. Find the 4 numbers. 7.03

b Sean collected \$1 and \$2 coins until he had 275 coins, having a total value of \$402. How many \$2 coins did Sean have?

36 Amit is at ground level 72 metres from the base of an apartment building. The angle of elevation to the top of the building is 35°. Find the height of the building, correct to the nearest metre. 4.08

37 Solve each equation. 7.05

a $5m^2 + 1 = 81$ **b** $\frac{x^2}{3} = 12$

PATH

38 Solve each equation. 7.07

a $2r^3 = 1024$ **b** $\frac{y^3}{6} = -121.5$

39 Solve each equation. 7.06

a $n^2 + n - 72 = 0$ **b** $u^2 - 5u - 66 = 0$ **c** $a^2 - 6a + 9 = 0$

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